Course Title:

BIT302 Software Engineering

Course Contents

Unit 9: Software Quality Assurance

- Introduction to Software Quality,
- Software Quality Assurance,
- Software Reviews

Questions:

1. What is Software quality management ? Explain Software quality management at an organizational level and project level.
2. Explain the differences between quality assurance and quality control.
3. What is Software quality planning? What are the outline structure for a quality planning?
4. What is Software standards? Explain its importance.
5. Explain Process and Product quality standards of Software quality management.

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Introduction to Software Quality Management:

- The key objective of software engineering is to produce on-time, high-quality software.
- Software quality management is concerned with ensuring that developed software systems are "fit for purpose." That is, systems should meet the needs of their users, should perform efficiently and reliably, and should be delivered on time and within budget.
- The use of quality management techniques along with new software technologies and testing methods has led to significant improvements in the level of software quality.
- Software Quality Management is concerned with ensuring that the required level of quality is achieved in a software product.

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Introduction to Software Quality Management:

Software quality management is both an organizational and an individual project issue:

1. At an organizational level, quality management is concerned with establishing a framework of organizational processes and standards that will lead to high-quality software. The QM team should take responsibility for defining the software development processes to be used and standards that should apply to the software and related documentation, including the system requirements, design, and code.
2. At a project level, quality management involves the application of specific quality processes, checking that these planned processes have been followed, and ensuring that the project outputs meet the defined project standards. Project quality management may also involve defining a quality plan for a project. The quality plan should set out the quality goals for the project and define what processes and standards are to be used.

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Quality Assurance Vs Quality Control

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Software Quality Management: Quality Assurance and Quality Control

Software quality management quality assurance and quality control are widely used.

- Quality assurance is the definition of processes and standards that should lead to high-quality products and the introduction of quality processes into the developing process.
- Quality control is the application of these quality processes to weed out products that are not of the required level of quality. Both quality assurance and quality control are part of quality management.

QA include the verification and validation and the processes of checking that the quality procedures have been properly applied.

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Quality management Vs Software Development

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Quality management activities

Software Development:

- Software development refers to the process of creating software applications through programming, testing, and maintenance. It involves designing, coding, testing, and deploying software systems.
- The main objective is to deliver functional, reliable, and efficient software that meets user requirements.
- Software development methodologies include Agile, Waterfall, DevOps, and others, each with its own approach to managing the development lifecycle.

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Quality management activities

Quality management:

- Quality management focuses on ensuring that products or services meet the desired quality standards. It encompasses processes, policies, and procedures aimed at consistently delivering products that meet customer requirements and expectations.
- The primary goal is to ensure that quality is built into every stage of the product lifecycle, from planning and development to deployment and support.
- Quality management involves methodologies such as Total Quality Management (TQM), Six Sigma, Lean, and various quality assurance techniques to monitor, measure, and improve processes.

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Software Quality: Quality management activities

- Quality management provides an independent check on the software development process.
- The quality management process checks the project deliverables to ensure that they are consistent with organizational standards and goals
- The QA team should be independent from the development team so that they can take an objective view of the software. This allows them to report on software quality without being influenced by software development issues.

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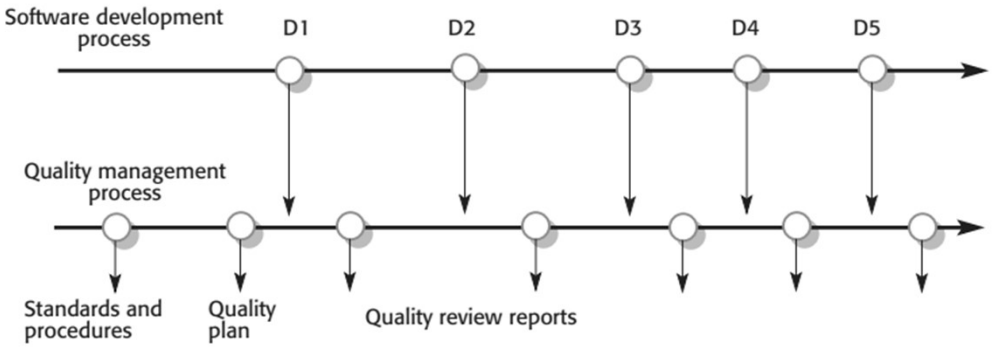
Software Quality: Quality management and software development



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Software Quality: Quality planning

- A quality plan sets out the desired product qualities and how these are assessed and defines the most significant quality attributes.
- The quality plan should define the quality assessment process.
- It should set out which organisational standards should be applied and, where necessary, define new standards to be used.

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Software Quality: Quality planning

An outline structure for a quality plan includes the following:

1. Product introduction A description of the product, its intended market, and the quality expectations for the product.
2. Product plans The critical release dates and responsibilities for the product, along with plans for distribution and product servicing.
3. Process descriptions The development and service processes and standards that should be used for product development and management.
4. Quality goals The quality goals and plans for the product, including an identification and justification of critical product quality attributes.
5. Risks and risk management The key risks that might affect product quality and the actions to be taken to address these risks

Quality plans should be short, concise documents If they are too long, no-one will read them.

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Software Quality: Software quality attributes

Therefore, software quality is not just about whether the software functionality has been correctly implemented, but also depends on non-functional system attributes. These attributes reflect the software dependability, usability, efficiency, and maintainability.

Safety	Understandability	Portability
Security	Testability	Usability
Reliability	Adaptability	Reusability
Resilience	Modularity	Efficiency
Robustness	Complexity	Learnability

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Software Quality: Quality conflicts

- It is not possible for any system to be optimized for all of these attributes – for example, improving robustness (not wholly affected by a single application failure) may lead to loss of performance.
- The quality plan should therefore define the most important quality attributes for the software that is being developed.
- The plan should also include a definition of the quality assessment process, an agreed way of assessing whether some quality, such as maintainability or robustness, is present in the product.

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Software Quality: Process and product quality

Two types of standards may be established as part of the quality assurance process:

- Product standards: document standards (structure), documentation standards (comment headers), and coding standards (variable names).
- Process standards define the processes that should be followed during software development (validation process).

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Software Quality: Process and product quality

- The quality of a developed product is influenced by the quality of the production process.
- This is important in software development as some product quality attributes are hard to assess.
- However, there is a very complex and poorly understood relationship between software processes and product quality.
  - The application of individual skills and experience is particularly important in software development;
  - External factors such as the novelty of an application or the need for an accelerated development schedule may impair product quality.

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Software standards:

- Standards define the required attributes of a product or process. They play an important role in quality management.
- Standards may be international, national, organizational or project standards.
- Product standards define characteristics that all software components should exhibit e.g. a common programming style.
- Process standards define how the software process should be enacted (sanctioned).

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Software standards: Importance of standards

- Encapsulation of best practice - avoids repetition of past mistakes.
- They are a framework for defining what quality means in a particular setting i.e. that organization's view of quality.
- They provide continuity - new staff can understand the organisation by understanding the standards that are used.

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Software standards: Product and process standards

Product standards	Process standards
Design review form	Design review conduct
Requirements document structure	Submission of new code for system building
Method header format	Version release process
Java programming style	Project plan approval process
Project plan format	Change control process
Change request form	Test recording process

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Software standards: ISO 9001 standards framework

- An international set of standards that can be used as a basis for developing quality management systems.
- ISO 9001, the most general of these standards, applies to organizations that design, develop and maintain products, including software.
- The ISO 9001 standard is a framework for developing software standards.
  - It sets out general quality principles, describes quality processes in general and lays out the organizational standards and procedures that should be defined. These should be documented in an organizational quality manual.

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Software standards: ISO 9001 certification

- Quality standards and procedures should be documented in an organisational quality manual.
- An external body may certify that an organisation's quality manual conforms to ISO 9000 standards.
- Some customers require suppliers to be ISO 9000 certified although the need for flexibility here is increasingly recognised.

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Reviews and inspections:

- A group examines part or all of a process or system and its documentation to find potential problems.
- Software or documents may be 'signed off' at a review which signifies that progress to the next development stage has been approved by management.
- There are different types of review with different objectives
  - Inspections for defect removal (product);
  - Reviews for progress assessment (product and process);
  - Quality reviews (product and standards).

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Reviews and inspections: Quality reviews

- A group of people carefully examine part or all of a software system and its associated documentation.
- Code, designs, specifications, test plans, standards, etc. can all be reviewed.
- Software or documents may be 'signed off' at a review which signifies that progress to the next development stage has been approved by management.

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Reviews and inspections: Program inspections

- These are peer reviews(examined by author's colleagues, in order to evaluate the work product's technical content and quality) where engineers examine the source of a system with the aim of discovering anomalies and defects.
- Inspections do not require execution of a system so may be used before implementation.
- They may be applied to any representation of the system (requirements, design, configuration data, test data, etc.).
- They have been shown to be an effective technique for discovering program errors.

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Reviews and inspections: Inspection checklists

- Checklist of common errors should be used to drive the inspection.
- Error checklists are programming language dependent and reflect the characteristic errors that are likely to arise in the language.
- In general, the 'weaker' the type checking, the larger the checklist.
- Examples: Initialization, Constant naming, loop termination, array bounds, etc.

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Key points:

- Software quality management is concerned with ensuring that software has a low number of defects and that it reaches the required standards of maintainability, reliability, portability and so on.
- SQM includes defining standards for processes and products and establishing processes to check that these standards have been followed.
- Software standards are important for quality assurance as they represent an identification of ‘best practice’.
- Quality management procedures may be documented in an organizational quality manual, based on the generic model for a quality manual suggested in the ISO 9001 standard.
- Reviews of the software process deliverables involve a team of people who check that quality standards are being followed.
- In a program inspection or peer review, a small team systematically checks the code. They read the code in detail and look for possible errors and omissions.